MIL-D-60093A (MU)
29 October 1973
SUPERSEDING
MIL-D-60093 (MU)
30 OCTOBER 1964 AND
MIL-D-60092 (MU)
30 OCTOBER 1964
(see 6.6)

MILITARY SPECIFICATION

DETONATOR, STAB, M50
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for, loading, assembling and packing for one type of detonator designated as Detonator, Stab, M50.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions

FSC: 1375

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MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8797796 - Detonator, M50, Assembly
8816367 - Firing Pin Guide
8816368 - Disc Lead
8827934 - Ball Steel
9200012 - Firing Pin
9200602 - Bushing
9200695 - Firing Assembly
9258182 - Box, Set-Up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set-Up, Packing, Ammunition, Outer for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator. - The detonators shall comply with all requirements specified on Drawing (dwg.) 8797796 and with all requirements specified in applicable specifications.

3.3 Moisture Content of Lead Azide. - The moisture content of the lead azide used in loading the detonators shall not exceed 0.50 percent.

3.4 Moisture Content of Primer Mixture. - The moisture content of the primer mixture used in loading the detonators shall not exceed 0.30 percent.

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3.5 Functioning

3.5.1 Output. - The detonators shall function, on the first attempt, under an applied energy of 11.85 inch ounces, and shall cause perforation of the lead disc to a diameter of not less than 0.375 inch minimum.

3.5.2 Waterproofness. - The detonators shall be waterproof as evidenced by normal functioning, as specified in 3.5.1, after being immersed in water maintained at 75 degrees Fahrenheit (°F) plus or minus 5°F, at a depth of 2 inches minimum for not less than 48 hours.

3.6 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.7 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate sugmitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1.c.) to a Government approved facility as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 complete sets of parts and sub-assemblies (i.e., 50 each of every component and every sub-assembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing

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known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government

4.2.2 Inspections to be Performed. - Components and sub-assemblies will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any component or sub-assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component or sub-assembly in the sample to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with the applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonator assemblies shall contain:

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a. Parts of one lot interfix number from one supplier.

b. Primer mixture from one batch or blending of batches which results in homogeneity of all primer mixture used in the loading of a lot.

c. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

d. RDX from not more than one lot.

4.3.2 Examination. - The examinations listed in the following sub-paragraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Disc, Detonator (see dwg. 9287495 covering a detail of dwg. 8797796).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			
Major:	AQL 0.40 Percent		
101.	Diameter	Gage	01001
102.	Thickness	Gage	01002

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	01003

4.3.2.2 Disc, Detonator Closing (see dwg. 9287496 covering a detail of dwg. 8797796).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Diameter	Gage	02001
102.	Thickness	Gage	02002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	02003

4.3.2.3 Cup, Detonator (see dwg. 9283131 covering a detail of dwg. 8797796).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Diameter of hole, min.	Gage	03001
102.	Concentricity of flash hole with outside diameter	Gage	03002
103.	Thickness through bottom	Gage	03003
104.	Length	Gage	03004
105.	Outside diameter	Gage	03005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	03006

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4.3.2.4 Assembly, Prior to Loading and Closing (see dwg. 8797796).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Interior coating incomplete or inadequate	Visual	04001
Major:	None defined.		
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	04002

4.3.2.5 Detonator Assembly (see dwg. 8797796).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup split or cracked	Visual	05001
Major:	AQL 1.00 Percent		
101.	Total length	Gage	05002
102.	Diameter	Gage	05003
103.	Color of disc improper	Visual	05004
104.	Not crimped full 360 degrees ..	Visual	05005
105.	Disc punctured, torn or loose .	Visual	05006
106.	Lacquer seal incomplete or inadequate	Visual	05007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	05008

4.3.2.6 Box, Set-Up, Inner, after loading and prior to closing slide cover (see dwg. 9258182).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Filler and cushion assembly missing	Visual	06001

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2.	Filler and cushion assembly improperly assembled	Visual	06002
3.	Stiffener missing	Visual	06003
4.	Slide cover assembled reversed.	Visual	06004
Major:	AQL 0.40 Percent		
101.	Release paper missing	Visual	06005
Minor:	AQL 1.00 Percent		
201.	Marking missing, incomplete or illegible	Visual	06006
202.	Evidence of poor workmanship (see 3.6)	Visual	06007

4.3.2.7 Box, Set-Up, Outer, with Inner set-up boxes assembled and prior to assembling top and sealing strip (see dwg. 9258183).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	07001
2.	Stiffener not facing one direction	Visual	07002
Major:	None defined.		
Minor:	AQL 100 Percent		
201.	Marking missing, incomplete or illegible	Visual	07003
202.	Evidence of poor workmanship (see 3.6)	Visual	07004

4.3.2.8 Box, Packing, Ammunition, prior to assembling cover and strapping (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.40 Percent		
101.	Bags containing set-up boxes damaged	Visual	08001
102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	08002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	08003

4.3.2.9 Box, Packing, Ammunition (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Strapping missing, broken, missing or improperly fastened	Visual-Manual	09001
102.	Box damaged	Visual	09002
103.	Marking missing, broken or illegible	Visual	09003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	09004

4.3.3 Testing

4.3.3.1 Moisture Content of Lead Azide (see 3.3) - Major Defect - Code No. 05009. - The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample shall be selected and tested as specified in 4.4.1. If the moisture content is in excess of the percentage specified and loading has not yet begun, the lot of lead azide shall be rejected. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

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4.3.3.2 Primer Mixture

4.3.3.2.1 Moisture Content (see 3.4) - Major Defect - Code No. 05010. - The moisture content of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample shall be selected and tested as specified in 4.4.2.1. If the moisture content is in excess of the percentage specified and if loading has not begun, the batch or blend of batches of the primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.3.3.2.2 Composition (see dwg. 8798896) - Major Defect - Code No. 05011. - The composition of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample in sufficient quantity to perform the test detailed in 4.4.2.2 shall be selected and tested as specified in 4.4.2.2. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing and if loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, detonators loaded with the questionable primer mixture shall be rejected.

4.3.3.3 Binder - Lubricant Content of RDX (see dwg. 8797796) Major Defect - Code No. 05012. - The binder-lubricant content of the RDX shall be determined for each lot at the time of loading. A sample of five (5) grams shall be selected from each lot and tested as specified in 4.4.3. If the binder-lubricant content is in excess of the percentage specified and if loading has begun, the lot of RDX shall be rejected until corrective action has been accomplished, as verified by repeating the binder-lubricant determination specified in 4.4.3. If loading of the detonators has begun, the determination shall be made on the unloaded RDX. If the binder-lubricant content of the unloaded RDX is in excess of the percentage specified, the unloaded RDX shall be rejected until corrective action has been accomplished and all detonators loaded with RDX from the rejected lot shall be rejected.

4.3.3.4 Functioning

4.3.3.4.1 Output (see 3.5.1) - Major Defect - Code No. 05013. - A sample of 315 detonators shall be selected at random from each lot. If three or more detonators fail to comply with the requirement specified, the lot shall be rejected. If one or two defects are found, a second sample of 315 shall be selected at random and tested. If, in the combined first and second sample, a total of four or more defects are found, the lot shall be

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rejected. The test shall be performed as specified in 4.4.4.1 using test equipment as specified in 4.3.4.

4.3.3.4.2 Waterproofness (see 3.5.2) - Major Defect - Code No. 05014. - A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirement specified, the lot shall be rejected. The test shall be performed as specified in 4.4.4.2 using test equipment specified in 4.3.4.

4.3.3.5 Heat Seal Test of Sealed Bag (see dwg. 9258183) - Major Defect - Code No. 07005. - Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.4 Test Methods and Procedures

4.4.1 Moisture Content of Lead Azide. - The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Method 101.5 except use a two gram sample and cover the dish with a sheet of polytetrafluoroethylene.

4.4.2 Primer Mixture

4.4.2.1 Moisture Content of Primer Mixture. - A sample of 0.7 grams shall be accurately weighed and dried to a constant weight at 60°C + 5°C or in a desiccator containing calcium chloride. The loss in weight shall be calculated as the percentage of moisture.

4.4.2.2 Composition of Primer Mixture

4.4.2.2.1 Potassium Chlorate

4.4.2.2.1.1 Preparation of Solutions

4.4.2.2.1.1.2 Solution No. 1. - To approximately 800 ml of distilled water in a 1-liter amber glass reagent bottle, 40 gms of potassium thiocyanate shall be added. The reagent bottle and

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contents shall be sealed with a rubber stopper and shaken until the solid material has been dissolved. To this solution, 1.0 gm of lead thiocyanate, 1.0 gm of antimony sulfide and 1.0 gm of lead azide shall be added. Again, the reagent bottle shall be stoppered, and the contents vigorously shaken for several minutes then allowed to stand over-night. Portions of the saturated solution shall be filtered before use.

4.4.2.2.1.3 Solution No. 2. - To approximately 400 ml of distilled water in a 1/2-liter amber glass reagent bottle, 1.0 gm of lead thiocyanate, 1.0 gm of antimony sulfide and 1.0 gm of lead azide shall be added. The reagent bottle shall be sealed with a rubber stopper and the centents vigorously shaken for several minutes then allowed to stand over-night. Portions of the saturated solution shall be filtered before use.

4.4.2.2.1.4 Procedure. - A portion of approximately 1.0 gm of the sample shall be accurately weighed directly into a tared 30-ml sintered glass crucible of medium porosity. From a buret, 3.0-ml of solution No. 1 shall be added to the crucible. The solution shall be swirled for exactly one (1) minute, care being taken that no liquid be spilled, then aspirated with suction and the bottom of the crucible wiped dry with tissue paper. The washing shall be continued, using one more 3.0-ml portion, two 2.0-ml portions and two 1.0-ml portions, making a total of 12.0-ml in all. The crucible shall be removed and the bottom of the crucible, the funnel, the adapter and the suction flask washed with water. A test tube shall be placed in the flask and the stem of the filtering funnel inserted into it. As above, the crucible and contents shall be washed with one 1.0-ml portion of solution No. 1. The solution shall be aspirated with suction and the filtrate caught in the test tube. The test tube shall be removed, 1.0-ml of distilled water added, then tested for the chlorate as follows:

The tube shall be inclined at an angle of about 45 degrees. About 0.5-ml of a solution containing 1.0 gm of diphenylamine in 100 gm of concentrated sulfuric acid shall be run down the side of the tube from a dropper so that there will be two distinct layers. If an appreciable amount of chlorate ion is present, a distinct blue ring shall be visible at the border of the two layers. On shaking the tube, the ring disappears but the contents assume a blue coloration which shall last several seconds, depending on the amount of chlorate ion present. If only a trace of chlorate ion is present, the blue ring may not form; but upon shaking the test tube, a slight bluish coloration shall appear momentarily throughout the solution and last for only a fraction of a second. If a definite blue ring forms, the washing shall be repeated as

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before, using one 1.0-ml portion of solution No. 1, and finally a 0.5-ml portion of the same solution, as needed, depending upon the outcome of the tests for the chlorate ion. (Usually a total of 14.0 or 14.5-ml of solution No. 1 shall be sufficient.) After the removal of potassium chlorate has been completed, the inside of the crucible shall be washed with a 1.0-ml of solution No. 2 and aspirated immediately. The bottom of the crucible shall be wiped dry with tissue paper, and the crucible dried in an oven at 80 degrees plus or minus 1 degree C for 30 minutes, cooled in a desiccator and weighed. The percentage of potassium chlorate shall be cauculated as follows:

$$\text{Potassium Chlorate, Percent} = \frac{B \times 100}{W}$$

where:

B = Loss in weight of the contents of the crucible.
W = Weight of the sample on a moisture-free basis.

4.4.2.2.2 Antimony Sulfide

4.4.2.2.2.1 Preparation of Solution of Ferric Ammonium Sulfate. - To approximately 120-ml of distilled water in a 1/2-liter reagent bottle, 125 gm of ferric ammonium sulfate shall be added. The reagent bottle shall be stoppered, vigorously shaken for several minutes, and then allowed to stand over-night. Portions of the saturated solution shall be filtered before use.

4.4.2.2.2.2 Procedure. - A portion of approximately 1 gm of the sample shall be accurately weighed directly into a tared 30-ml sintered glass crucible of medium porosity. A 5.0-ml portion of one (1) percent nitric acid shall be added and the crucible constantly swirled for one (1) minute, in order that all the ingredients of the powder except antimony sulfide by dissolved. Care shall be taken that none of the liquid is spilled. The mixture shall be filtered by applying suction, employing a vacuum flask. The crucible shall be removed from the adapter and the bottom wiped dry with tissue paper. The washing operation shall be repeated using 5.0-ml of one (1) percent nitric acid each time, and the contents swirled for exactly one (1) minute. After a total of 40-ml of one percent nitric acid has been used, the crucible shall be removed and the bottom of it washed with water. The funnel, the funnel adapter and the vacuum flask shall be washed with water and the filtrate and washings discarded. About 1-ml of a saturated solution of ferric

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ammonium sulfate shall be placed in the flask. The contents of the crucible shall be treated as before with 5.0-ml of one percent nitric acid, and filtered by means of suction into the flask. If the liquid in the flask turns red due to the presence of the sulfocyanate ion in the filtrate, the washing of the contents in the crucible shall be repeated with additional 5.0-ml portions of one percent nitric acid until a negative test for the thiocyanate has been affected. (Usually a total of 50-ml shall be sufficient.) The crucible and contents shall be rinsed with alcohol, and then with ether. The solution shall be aspirated until the odor or ether can no longer be detected. The crucible and contents shall be dried in an oven at 80 degrees plus or minus one (1) degree C for 30 minutes, cooled in a desiccator and weighed. The percentage of antimony sulfide shall be calculated as follows:

$$\text{Antimony Sulfide, Percent} = \frac{A \times 100}{B}$$

where:

A = Weight of residue in the crucible.
B = Weight of sample on a moisture-free basis.

4.4.2.2.3 Lead Azide. - The lead azide content shall be determined using the (Eudiometer) method and apparatus in MIL-STD-650, Method No. 407.1.

4.4.2.2.4 Lead Thiocyanate. - The percentage of lead thiocyanate in the sample shall be calculated on a moisture-free basis by subtracting from 100 the sum of percentages of potassium chlorate, antimony sulfide and lead azide.

4.4.3 Determination of Binder-Lubricant Content of RDX. - The calcium resinate, graphite and total binder-lubricant content shall be determined as follows:

4.4.3.1 Calcium Resinate. - An accurately weighed portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three (3) 20-milliliter (ml) portions of chloroform which have been previously saturated with RDX, maintaining each portion in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up by stirring occasionally and the sides of the crucible washed down with each 20-ml portion of chloroform to insure complete removal of the calcium resinate. The residue shall

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be aspirated until the odor of chloroform is no longer detectable; the crucible and contents shall be placed in a drying oven at 95 degrees C plus or minus 2 degrees C for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying and weighing with individual 20-ml portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 grams. The loss in weight of the crucible plus contents shall be calculated to percent resinate in the sample on a moisture-free basis as follows:

$$\text{Percentage of Calcium Resinate} = \frac{100A}{W}$$

where:

A = Loss in weight of the crucible plus contents.
B = Weight of the sample.

4.4.3.2 Graphite. - The residue remaining after the calcium resinate has been removed shall be extracted with two (2) 20-ml portions of dimethyl formamide, each portion maintained in contact with the residue for approximately five (5) minutes with intermittent stirring before applying suction. The crucible and contents shall be washed with a minimum of 3-ml portions of acetone, each portion maintained in contact with the residue for approximately one (1) minute with continuous stirring before applying suction, until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 25-ml beaker, evaporated to dryness on a steam bath and several drops of a solution of chromotropic acid added in 36N sulfuric acid. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner, indicates the presence of RDX. If RDX is present, continue the acetone washing until a negative test is obtained. The residue shall be aspirated until the odor of acetone is no longer detectable; the crucible and contents shall be placed in a drying oven at 95 degrees C plus or minus two (2) degrees C for 1/2 hour, cooled in a desiccator and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture-free basis as follows:

$$\text{Percentage of Graphite} = \frac{100A}{W}$$

where:

A = Increase in weight of filtering crucible.
W = Weight of sample.

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4.4.4 Functioning

4.4.4.1 Output. - The detonator shall be assembled to the applicable te st fixture specified in 4.3.4 so that the firing pin will strike the sensitive end of the detonator. A steel ball weighing 3.95 ounces maximum shall be dropped on the firing pin to function the detonator from a height of three (3) inches maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. The Method of inspection shall be in accordance with dwgs. 8816367, 8816368, 8827934, 9200012, 9200602 and 9200695.

4.4.4.2 Waterproofness. - The detonator shall be immersed under water as specified for the time specified. Within one (1) hour after removal from the water, the detonator shall be subjected to the output test specified in 4.4.4.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C. - Packaging shall be as specified on dwgs. 9258182 and 9258183.

5.2 Packing

5.2.1 Level C. - Packing shall be as specified on dwg. 9258184.

5.3 Marking. - Marking of all packaging and packing shall comply with dwgs. 9258182, 9258183 and 9258184.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

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6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific

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identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of Test Data and Ammunition Data Cards. - In addition to the normal distribution of records for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commander
U.S. Army Picatinny Arsenal
ATTN: SARPA-QA-A-M
SARPA-AD-E-P-2
Dover, New Jersey 07801

6.5 Check Test for Possible Deterioration. - The following test is designed to prevent major ammunition items being loaded with detonators that may have deteriorated. This test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require a check test, the Contracting Officer shall insert into the contract a check test performance provision. The Contracting Officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the deterioration check test.

6.5.1 Functioning. - A sample of 315 detonators shall be selected at random from each lot. If four (4) or more detonators fail to comply with the requirement specified, the lot

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shall be rejected. If more than one defect is found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of five (5) or more defects are found, the lot shall be rejected. The test shall be performed as specified in 4.4.4 using test equipment as specified in 4.3.4.

6.6 Supersession Data. - This specification includes the requirements of MIL-D-60092 (MU) dated 30 October 1964.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1375-A-038

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 **1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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